

M. Sc. Physics (Semester-II) 2026

Quantum Mechanics II – Tutorial 2

1. (a) Calculate the probability that a hydrogen $1s$ electron will be found within a distance λ from the nucleus. The wave function for the $1s$ orbital of hydrogen is

$$\psi_{100} = \frac{1}{\sqrt{\pi}} \left(\frac{1}{a_0} \right)^{3/2} e^{-r/a_0}.$$

- (b) What is the radius of the sphere that encloses a 90% probability of finding a hydrogen $1s$ electron.
2. The energy of the bound state of Hydrogen is given by

$$E_n = \frac{-13.6eV}{n^2}.$$

Calculate the degeneracy of the n^{th} eigenstate.

3. (a) Write down the orthogonality relation for spherical harmonics $Y_\ell^m(\theta, \phi)$.
(b) Why are spherical harmonics orthogonal?
(c) Why is there a factor of $\sin \theta$ when we integrate in (a).
4. For the angular momentum quantum number $\ell = 3$, find the representations of the operators L^2 , L_z , L_+ , and L_- in the $|\ell, m\rangle$. Use the standard action of these operators on angular momentum states.